$^{1}H(^{34}Ar,P')$ 2001Kh17

2001Kh17: a 47-MeV/nucleon ^{34}Ar was produced via the projectile fragmentation of a 95-MeV/nucleon ^{36}Ar primary beam impinging on a ^{12}C target at GANIL. The secondary target was CH_2. The elastic and inelastic protons were detected using the MUST Si-Si(Li)-CsI telescope and heavy nuclei were analyzed by the SPEC spectrometer. Measured $\sigma(E_p,\theta)$. Deduced levels, J, π, and deformation parameters from Coupled-Channel Born Approximation (CCBA) using the ECIS code and TAMURA-DWBA analyses of the measured $\sigma(\theta)$. Also see 2000KhZX, 2001Bl17, 2003La13 for more discussions on the same data.

^{34}Ar Levels

E(level)	J^{π}†	L	Comments
0	0^+	0	Ratio of quadrupole matrix elements $M_n/M_p/(N/Z)$=1.35 *28* from phenomenological analysis (2001Kh17) and 0.99 *17* from the QRPA transition densities normalized to the data in 2001Kh17.
1.9×10^3 *2*	2^+	2	β_2=0.27 *2* (2001Kh17)
4.5×10^3 *2*	3^-	3	β_3=0.39 *3* (2001Kh17)

† From CCBA and DWBA analyses of measured $\sigma(\theta)$ in 2001Kh17.